

OPERATION ROTOR TRACE

Case Story

On **18 August 2026**, a Bell 412 helicopter operated by **Xavier Air Logistics** departed Accra on a scheduled flight to Tarkwa. Approximately 38 minutes after take-off, the helicopter disappeared from air-traffic monitoring and crashed in a remote forested area along the flight route.

Before the crash, the pilot, **Captain Daniel Mensah**, reported intermittent navigation problems but did not immediately declare an emergency. During the final minutes of the flight, cockpit recordings captured the crew discussing increasing gearbox temperature, decreasing oil pressure and unusual vibration. Witnesses near the crash location later reported hearing an abnormal mechanical sound before seeing the helicopter descend rapidly.

The preliminary investigation revealed that the helicopter had undergone maintenance three days before the accident. A replacement gearbox oil-pressure sensor assembly had been installed, but investigators discovered a possible discrepancy between the serial number on the installed component and the number recorded on its supporting certificate.

Further examination uncovered two versions of a maintenance report. The original report described the aircraft as **AIRCRAFT NOT SERVICEABLE**, while a later version stated **CLEARED FOR FLIGHT**. Investigators suspect that portions of the electronic maintenance history may have been modified after the original engineering assessment.

A USB storage device recovered from the maintenance office contained maintenance documents, diagnostic exports, component certificates, invoices and photographs. The maintenance laptop used by engineer **Kwame Adu** also contained browser searches relating to deleting diagnostic logs, removing document metadata and clearing maintenance-system history.

Company communications revealed that Operations Manager **Isaac Tetteh** had been pressing the engineering team for confirmation that the aircraft would be available for the Tarkwa flight. Stores and Procurement Officer **Linda Agyeman** was linked to the acquisition and documentation of the replacement component, while Senior Engineer **Emmanuel Quartey** had previously discussed recurring gearbox warnings with the maintenance team.

The pilot's phone contained calls and messages involving the maintenance and operations personnel. One message warned that the gearbox indication had returned during a ground run and advised the pilot to monitor the temperature. Another message later informed him that the paperwork had been cleared and the aircraft released.

Digital evidence recovered from the helicopter includes the **Flight Data Recorder/black box, Cockpit Voice Recorder, HUMS data, GPS records, flight telemetry and the flight tablet**. These sources show the aircraft's actual route, altitude, heading, vibration, gearbox temperature, oil pressure, rotor RPM and other system parameters.

The helicopter also deviated from its original flight route. However, weather reports indicate heavy cloud and rainfall along part of the planned route, suggesting that the deviation may have been operationally reasonable rather than suspicious.

You have been engaged to examine the recovered digital evidence, reconstruct the sequence of events, determine whether maintenance records were altered, establish what the crew knew before and during the flight, and identify whether the digital evidence supports maintenance negligence, record falsification, component irregularities, mechanical failure, or another explanation.

Students must support every conclusion with appropriate forensic artifacts, timestamps and corroborating evidence.

Investigation Questions

- 1. What was the original maintenance status of the Bell 412 before the flight, and what forensic evidence shows that the report was later changed to “CLEARED FOR FLIGHT”?**
- 2. Who modified the maintenance report, when was it modified, and which digital artifacts can be used to attribute the change to a particular user or device?**
- 3. What serial number appears on the replacement component installed on the helicopter, and does it match the component certificate, invoice and maintenance records?**
- 4. What suspicious browser searches were performed on the maintenance laptop, and what do their timestamps reveal about possible attempts to delete or conceal maintenance evidence?**
- 5. What does the recovered USB evidence reveal about the original maintenance report, diagnostic exports, component documentation and deleted files, and when was the USB connected to the maintenance laptop?**
- 6. Using the telemetry, HUMS and black-box/FDR data, reconstruct the sequence in which gearbox temperature, oil pressure and vibration changed during the final stage of the flight. Which abnormal condition appeared first?**
- 7. What did Captain Daniel Mensah and the other crew member say during the final minutes of the Cockpit Voice Recorder, and how does their conversation correlate with the FDR, HUMS and telemetry data?**
- 8. What communications occurred among Captain Daniel Mensah, Kwame Adu, Isaac Tetteh, Emmanuel Quartey and Linda Agyeman before the flight, and which communications are most significant to the investigation?**
- 9. Was the helicopter's route deviation itself suspicious? Compare the planned route, actual GPS/FDR track, ATC communications and weather evidence to justify your conclusion.**
- 10. Based on all available digital evidence, construct a final timeline from the maintenance work three days before the crash through the final recorded flight event, and provide your forensic assessment of the most significant factors that contributed to the incident.**

